

Lab 2 Report

CSE 60685 Machine Learning for Embedded Systems LeNet Architecture Comparison on Raspberry Pi

Device and settings. Supplied Raspberry Pi 4B Rev. 1.5; Raspberry Pi OS 64-bit (Debian 13), Python 3.13.5, PyTorch 2.8.0+cpu. All training, evaluation and timing ran on the Pi CPU with one intra-op and one inter-op thread. Same CanaKit power supply, closed case, running fan and ondemand governor; sequential runs. Pre-warmup benchmark temperatures were 42.8–44.3 °C; all recorded power/throttling flags were 0x0.

Common recipe. Fashion-MNIST: supplied 12,000/2,000 training/validation split and all 10,000 official test images; grayscale 28×28 float32 inputs in $[0, 1]$, no extra normalization or augmentation. Each design was fixed before test evaluation and trained from scratch: 5 epochs, batch 64, Adam, learning rate 0.001, seed 42, zero loader workers. Final-epoch checkpoints were reloaded and verified; A also passed a separate verify-only process before test evaluation. C uses hidden widths **60 and 42**.

Model	Parameters	Test accuracy (%)	Mean training (s/epoch)	Median inference (ms)
A: baseline	44,426	79.72	13.86	2.6603
B: conv-small	27,180	78.05	10.17	2.3970
C: FC-small	20,984	78.84	13.56	2.4440

Timing. Unchanged instructor `train.py`, `evaluate.py`, `benchmark.py` and `summarize.py`. Training time is the mean of all five training loops, including data iteration, forward/backward, updates and fixed statistics; validation, test and output are excluded. After all training, inference used the same prepared $[1, 1, 28, 28]$ tensor, 10 untimed warmups and 100 timed calls per model in evaluation/inference mode. The timer covers synchronous `model(tensor)` plus Python call overhead, excluding loading, preprocessing, `argmax/softmax` and output.

1. Architecture changes

A uses convolution channels 6/16 and hidden widths 120/84. B changes the channels to 3/8, retaining 120/84. For C, I chose to halve both hidden widths to 60/42 while keeping channels 6/16. The second pooling layer produces $c_2 \times 4 \times 4$ features, so the first Linear layer needs $16c_2$ inputs: 256 for A/C and 128 for B. Kernels, pooling and activations are unchanged.

2. Parameter count and measured speed

C has the largest parameter reduction: **52.77%**, versus 38.82% for B. B has the largest measured inference speedup: **1.11×** versus A, compared with 1.09× for C (baseline median divided by candidate median). This supports my interpretation that architecture matters, not just parameter count. C removes weights from the final classifier but keeps the convolutional feature maps and their repeated spatial computation. B reduces those channels as well as the first Linear input. In this run, B also trained faster (10.17 versus 13.86 s/epoch); C's training time remained closer to A (13.56 s/epoch). Aggregate timings do not isolate each layer's contribution.

3. Model choice

I would choose **C** because my priority is fewer parameters and the accuracy differences here are small enough for that tradeoff. C uses 20,984 parameters, with test accuracy 0.88 percentage points below A and 0.79 points above B. Its inference median is lower than A's and only 0.0470 ms higher than B's. Thus, I prefer C's smaller weight-storage requirement while accepting this measured accuracy/runtime tradeoff. These are results from one training run per design, not evidence of statistical equivalence.

AI Attribution Appendix

Lab 2 — excluded from the one-page analysis limit

Tool. OpenAI Codex.

Main prompts (translated/summarized): “Explain Lab 2 step by step and help me complete it”; “Are A, B and C compliant with the requirements?”; “Make the attribution appendix more concise.”

Assistance. Codex helped interpret the instructions, set up the Pi through SSH, transcribe Tutorial Steps 2–3 into `models.py` and `train_step.py`, and execute the supplied training, evaluation and benchmark commands. It wrote separate helpers for sequential execution, cooling, logging and file verification, and assisted with result interpretation, English wording and LaTeX formatting. The instructor’s reference experiment scripts remained unchanged.

Student decisions and evidence. I selected C’s hidden widths of 60/42 and chose C after considering the measured tradeoff, prioritizing fewer parameters while accepting the accuracy difference. I explained that C preserves convolutional computation while shrinking the fully connected classifier. All reported measurements came from the actual Raspberry Pi. Checkpoints, original JSON/CSV results and the detailed assistance log are retained with the code.